Score-based Generative Modeling through Stochastic Differential Equations

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Chapter 1

Introduction

This English edition translates the mathematical exposition of the Spanish monograph. The document presents the forward diffusion, its density evolution, the reverse-time equation, score matching, and a one-dimensional numerical demonstration. The complete chapter-by-chapter translation is being checked against the original source before publication.

Chapter 2

Diffusion models as generative models

We start from samples of an unknown distribution p_0, perturb them with an ItÃt̃ stochastic differential equation, and learn the score field needed to reverse the process.

Chapter 3

Numerical demonstration

The accompanying scripts diffuse a shifted Gamma distribution with an Ornstein–Uhlenbeck process, learn its score with denoising score matching, and recover samples from the reverse process.

Chapter 4

Source and reproducibility

The Spanish source, figures, scripts, references, and the English poster are included in this repository. The numerical example is intentionally one-dimensional so that every step remains inspectable.